

```

Program Question_1
  Implicit None
  Integer::n,i,max_n=70
  Integer(8)::M
  Logical::is_prime
  Write(*,'(A)') "n          M_n      Prime"
  Write(*,'(A)') "_____ "

  Do n=0,max_n-1
    M=2**n-1
    is_prime=.True.
    If(M<2)Then
      is_prime=.False.
    Else
      Do i=2,int(Sqrt(real(M)))
        If(Mod(M,i)==0)Then
          is_prime=.False.
          Exit
        End If
      End Do
    End If
    Write(*,'(I3, T10, I20, T35, A)')n,M,merge('Yes','No ',is_prime)
  End Do
End Program

```